
Pocket-sensitive somatic variant effect prediction (VEP) using AlphaFold-predicted protein structures

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Abstract

Most missense variant effect predictors (VEPs) rely on sequence and evolutionary conservation, with 3D structural context entering only implicitly through learned representations. Driver variants are not spatially random. They cluster at functional sites like binding pockets and modulatory sites, where a model that sees a residue’s spatial relationship to the pocket has access to information that the residue-level sequence cannot recover. We test whether explicit pocket-aware features carry signal beyond what sequence and conservation already encode with a three-way orthogonal ablation across sequence, structure (AlphaFold + fpocket + DSSP), and evolution (phyloP100way, phastCons100way) feature blocks on 30,620 labeled ClinVar missense variants spanning 1,040 genes. Three classifiers (random forest, MLP, XGBoost) are trained on each single-modality arm, every pairwise combination, and the full union, swept across five gene-grouped seeds. The union beats every constituent in every model (macro F1 0.85–0.86 vs. best single-modality 0.74–0.78), and every pairwise combination also beats its best constituent. Structural and evolutionary features carry information that sequence-only features do not recover.

1 Background

Cancer treatment regularly incorporates tumor sequencing and other biomarker assays [1, 2] to identify disease-defining and actionable alterations among the thousands to hundreds of thousands of somatic mutations a tumor can accumulate [3]. OncoKB, an FDA-recognized¹ precision oncology knowledge base maintained by Memorial Sloan Kettering Cancer Center and canonical source in the cancer genomics community [4], curates 155 somatic driver alterations across 77 genes that have matched targeted therapies in routine clinical use², including BRAF V600E in melanoma (dabrafenib + trametinib), EGFR exon 19 deletions in NSCLC (osimertinib), KRAS G12C in NSCLC and colorectal cancer (sotorasib, adagrasib), and HER2 amplification in breast and gastric cancers (trastuzumab and HER2-directed ADCs). Targeted drugs made up 91% of FDA oncology approvals between 2000 and 2022 [5], and basket trials increasingly enroll patients on the basis of a shared molecular feature regardless of tumor origin [6], with programs like KEYNOTE-158 [7] serving as a proof of concept for tissue-agnostic FDA approvals.

However, that 77-gene actionable set is a fraction of the broader driver landscape. The rest is a long tail of less common variants, most of which are not yet functionally characterized or therapeutically

¹First somatic-variant knowledge base granted U.S. FDA partial recognition (October 2021); see the FDA list of recognized human genetic variant databases at <https://www.fda.gov/medical-devices/precision-medicine/fda-recognition-public-human-genetic-variant-databases>.

²OncoKB Actionable Genes table (<https://www.oncokb.org/actionable-genes>), accessed 2026-05-13.

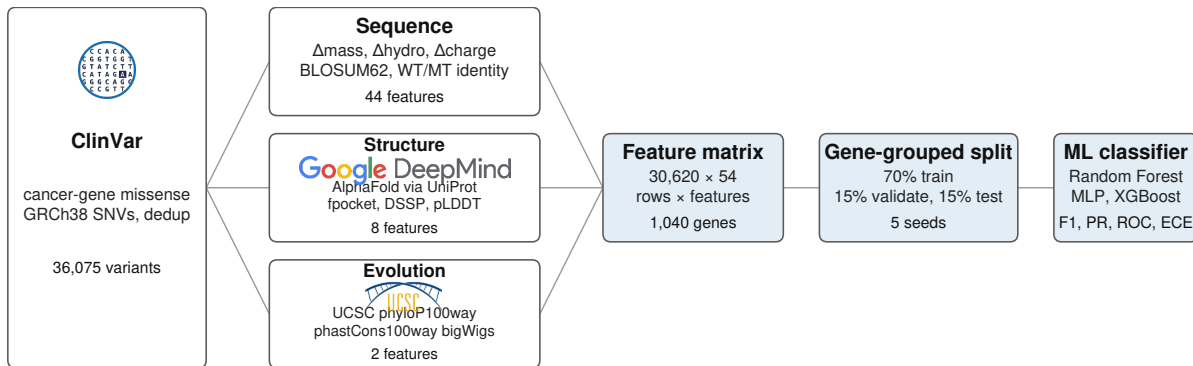


Figure 1: End-to-end pipeline. Variant labels are resourced from ClinVar and bridged to AlphaFold structures via UniProt; sequence, structure (fpocket + mkdssp + pLDDT), and evolutionary (phyloP / phastCons) feature blocks are joined into a single feature matrix, then evaluated under a gene-grouped split.

actionable. Kinnersley et al. [3] reported 74 driver genes new to any cancer in their 100kGP cohort, with 88% of those appearing at frequencies below 10% in their respective tumor types, and 79% of the 330 drivers in that cohort have no therapeutic target annotation in COSMIC or OncoKB. The bottleneck for growing the actionable driver catalog is variant interpretation, not data generation. Computational variant effect prediction is the nomination step in that workflow, and it is the focus of this paper.

Variant nomination tools have gone through several generations. Alignment-based scoring such as SIFT (2001) [8] and PolyPhen-2 (2010) [9] gave way to conservation-derived scores like phyloP (2010) [10], phastCons (2005) [11], and GERP (2005) [12], and to ensemble combiners like CADD (2014) [13] and REVEL (2016) [14]. A cancer-specific lineage including CHASM (2009) [15], FATHMM-cancer (2013) [16], and CanDrA (2013) [17] trained directly on somatic data. Deep generative models on protein-family alignments came next, with EVE (2021) [18] building on EVmutation (2017) [19] and DeepSequence (2018) [20], and PopEVE (2025) [21] extending the family by folding in population frequency. Protein language models like ESM (2021, 2023) [22, 23] followed, learning sequence statistics over broader protein universes without explicit family alignment. Through all of this, explicit 3D context has been hard to use systematically. Structure-aware predictors like SuSPect (2014) [24] and the 3D hotspot analyses of Gao et al. [25] (2017) existed but were limited by patchy experimentally resolved structures.

AlphaFold (2021, 2022) [26, 27] mitigated this constraint by providing high-confidence predicted structures for nearly the entire human proteome, and pocket-detection tools like fpocket (2009) [28] and P2Rank (2018) [29] could now run on any protein. The current generation of VEPs has folded structural information into the modeling pipeline, though almost always implicitly. AlphaMissense (2023) [30] adapts an AlphaFold-style sequence-to-structure architecture and trains on weak labels derived from population allele frequency, PrimateAI-3D (2023) [31] uses a 3D convolution over predicted neighborhoods, and ESM-family language models (2021) [32] pick up structural regularities through pretraining without explicit structural input. Because these models report a single combined score, it is hard to tell which signal class is doing the work, and so it stays unclear whether explicit pocket-level features add anything implicit structural reasoning does not already capture.

We address that question with a controlled feature ablation on somatic missense variants labeled as oncogenic or benign, featurizing each variant along three orthogonal blocks (sequence, structure, and evolution) and comparing every single-modality arm, every pairwise combination, and the union under a gene-grouped train/val/test split repeated across five seeds. This is a diagnostic ablation, not a new state-of-the-art predictor, and we do not benchmark against AlphaMissense. The question is whether explicit pocket conformation carries information that current implicit structural reasoning misses.

2 Motivations & Hypotheses

Driver variants are not spatially random. They cluster at residues that regulate enzymatic activity, occupy interface positions, or sit inside binding pockets that are themselves under selective pressure to remain conserved. A model that sees a residue’s pocket membership, distance to the nearest pocket, and the druggability score of that pocket has access to information a residue-only sequence model cannot reconstruct without implicitly redoing structure prediction. The ablation in this paper tests whether that information is recoverable in practice and whether it adds to sequence and evolutionary conservation rather than overlapping with them. We expect structure features on their own to carry classification signal above the prevalence baseline, every pairwise combination of sequence, structure, and evolution to beat its better constituent modality, and the union of all three to beat every pairwise combination.

3 Methods

3.1 Dataset

In the proposal we wanted to use OncoKB, which is the gold standard for querying cancer pathogenicity and drugability labels. On review, their terms of use bar reusing oncogenicity annotations as training labels for machine learning models in both academic and commercial settings. We instead resorted to ClinVar, which is not as curated. Its `variant_summary` dump is fully public and carries no such restriction, and it has both an explicit oncogenicity column (449 variants in our final set) and a broader clinical-significance column that we gated to OncoKB-curated cancer genes (gene names only, not labels) to avoid pulling in hereditary-disease variants. The oncogenicity-sourced rows drop from 449 to 434 after the inner join against feature coverage. The raw dump has ~ 8.9 M rows; after restricting to GRCh38 single-nucleotide missense substitutions and collapsing codon-degenerate duplicates at the (GeneSymbol, protein_change) level, we kept 36,075 labeled rows spanning 1,073 genes [33].

For structural context we used the AlphaFold human proteome (20,504 monomer PDB files, v4) [26, 27]. ClinVar gene symbols were mapped to UniProt accessions through UniProt’s mapping service, yielding 16,620 unique canonical accessions. Per-residue structural features were computed from each predicted structure with `fpocket` [28] and `mkdssp` [34].

For evolutionary context we used the UCSC `phyloP100way` and `phastCons100way` bigWig tracks, which score nucleotide-level conservation across a 100-vertebrate multiple alignment [10, 11]. The proposal originally specified ConSurf-DB; we switched to `phyloP`/`phastCons` because they are the canonical evolutionary signal used by REVEL, CADD, and dbNSFP, and because ClinVar already provides the GRCh38 locus and so the join is direct and per-nucleotide. ConSurf-DB remains available as a structure-aligned follow-up signal.

The full feature matrix is built by inner-joining all three feature blocks to the labeled set on a per-variant key, yielding 30,620 rows across 1,040 genes with full feature coverage and a 1.47:1 benign:oncogenic ratio.

3.2 Feature Engineering

Features split into three orthogonal blocks. The *sequence* block holds per-substitution physicochemical deltas (mass, Kyte–Doolittle hydrophathy, net charge at pH 7), the BLOSUM62 substitution score, and one-hot encodings of the WT and MT amino acid identities. The *structure* block holds `fpocket`-derived pocket geometry (distance to nearest pocket, in-pocket flag, druggability), `mkdssp`-derived solvent-accessible surface area and a 3-state secondary-structure encoding, and the AlphaFold per-residue confidence (pLDDT); pLDDT lives here rather than under sequence because it is a structural-quality signal, and treating it as sequence-level would muddy the orthogonality of the ablation. The *evolution* block holds UCSC `phyloP100way` and `phastCons100way` queried at the variant nucleotide. Per-column definitions and sources are in Appendix B.

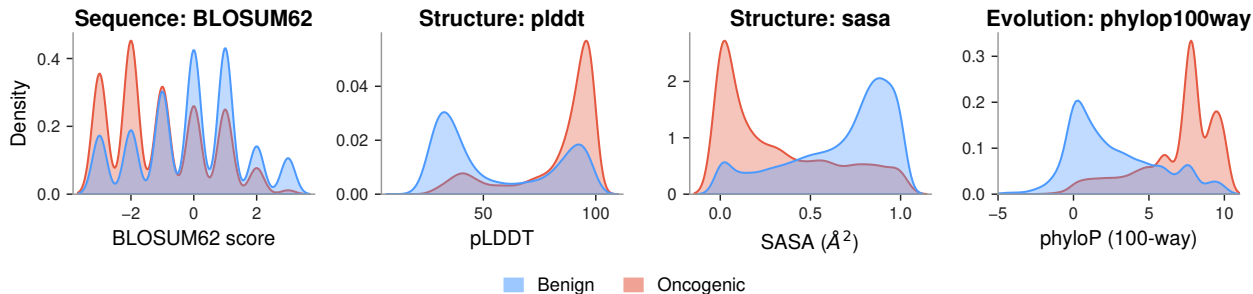


Figure 2: Per-feature distributions across the three feature blocks (sequence, structure, evolution), split by oncogenicity label. Kernel density estimates with class-wise normalization.

3.3 Data Exploration

Figure 2 shows a subset of features from all three blocks (sequence, structure, evolution). BLOSUM62 separates the two classes somewhat, with oncogenic variants skewing negative and benign variants skewing positive or zero, though the two distributions still overlap. The structure and evolution features separate the classes more cleanly. pLDDT and SASA (solvent-accessible surface area) both show visibly different peaks between benign and oncogenic variants, and phyloP100way behaves similarly. On these distributions alone the structure and evolution blocks look like the stronger candidates going into the model.

3.4 Model Choice & Evaluation

We evaluate three model families spanning bagged trees, boosted trees, and a small neural network, and so the ablation isolates the effect of the feature blocks rather than any single inductive bias. For the random forest we use sklearn’s RandomForestClassifier with `n_estimators=300`, unbounded depth, `min_samples_leaf=2`, and `class_weight="balanced"` [35]. For the MLP we use sklearn’s MLPClassifier wrapped in a StandardScaler then MLP pipeline (since the input columns mix 0/1 binary columns, pLDDT in $[0, 100]$, and unbounded floats); the MLP has two hidden layers of size 32 and 16, `max_iter=1000`, and `early_stopping=True` with `n_iter_no_change=20` [35]. For XGBoost we use XGBClassifier with `n_estimators=300`, `max_depth=4`, learning rate 0.1, log-loss eval metric, and `scale_pos_weight` set per split to the empirical $n_{\text{neg}}/n_{\text{pos}}$ ratio of the training fold (≈ 1.47) [36].

3.5 Class Distribution and Splits

The labels are mildly imbalanced at $\approx 1.47:1$ benign:oncogenic. The random forest is calibrated for this through `class_weight="balanced"` and XGBoost through `scale_pos_weight` set per split to the training-fold $n_{\text{neg}}/n_{\text{pos}}$. sklearn’s MLP has no `class_weight` parameter and we fit it unweighted; this leaves the MLP at a small disadvantage on the most imbalanced single-modality arms (visible as the larger gap between MLP and tree-based models on the sequence-only setting) but does not affect the ablation conclusion, which is driven by feature-set deltas an order of magnitude larger than the model-vs-model spread.

Splits are gene-grouped (sklearn GroupShuffleSplit on GeneSymbol) at 70 / 15 / 15 train / val / test; this is so that no protein reappears in both training and test evaluation. This is critical here because variants within a single oncogene tend to share a label, and without grouping the model can learn a gene-identity shortcut that inflates test metrics beyond what the features actually provide. Each of the five seeds redraws the gene partition and refits the models; per-split label balance is printed at runtime and so, any drift from the overall 1.47:1 ratio is visible.

4 Results

A full table of the results for the 7 feature set combinations and 3 models can be found in the Appendix (Table 1).

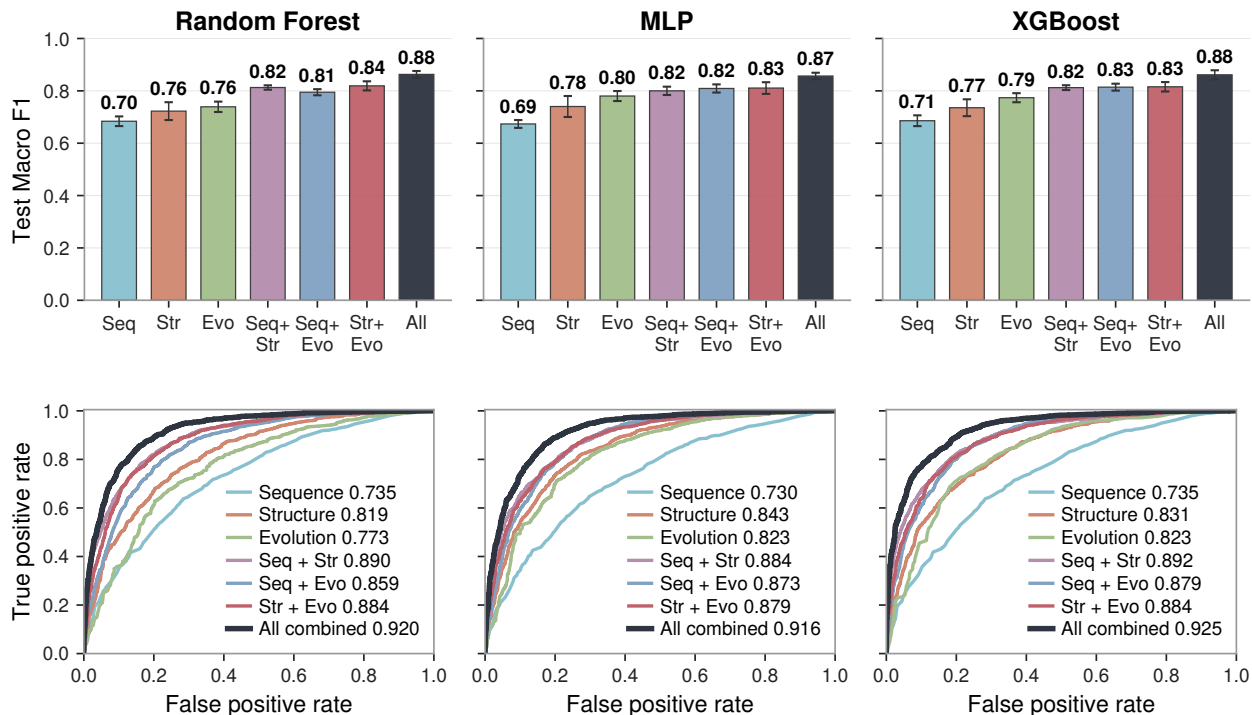


Figure 3: (Top) Test macro-F1 across the seven feature configurations, faceted by classifier. Error bars span ± 1 std over 5 gene-grouped seeds. **(Bottom)** Test-set ROC curves for the same seven configurations, with one panel per classifier. Bars and curves use a shared color palette so the same color traces a feature set from its macro-F1 bar to its ROC curve directly below.

4.1 Relative Model Performance

The full union of sequence, structure, and evolutionary features achieves the best macro-F1 in every model (Random Forest 0.863 ± 0.013 , MLP 0.857 ± 0.013 , XGBoost 0.861 ± 0.018 on test) and dominates every single-modality and pairwise arm. Among single-modality arms the ordering is consistent, with evolution beating structure beating sequence and a spread of ≈ 0.10 macro-F1 from worst to best (0.68 for sequence, 0.74 for structure, 0.78 for evolution). The seed-to-seed standard deviation also shrinks monotonically with feature richness, from $\sigma \approx 0.03$ on the structure-only arm to $\sigma \approx 0.014$ on the union, so richer feature sets yield more stable training as well as higher mean performance. Beyond raw macro-F1, every pairwise combination beats its strongest single-modality constituent (+0.03 to +0.09 macro-F1) and the union beats its strongest single (+0.08 to +0.12). No one modality is carrying the headline number with the others contributing noise. This matches what we expected for structure features carrying signal on their own and for the union beating every pairwise combination.

5 Discussion

Permutation importance on the union arm agrees across all three models on which features are contributing most. The top four features by importance are, in order, phylo100way, plddt, sasa, and BLOSUM62; phastcons100way and the pocket features sit in the next tier. The strongest single feature is evolutionary conservation (consistent with the broader missense-VEP literature), but the second- and third-strongest are both structural, and crucially neither of them comes from fpocket. plddt is AlphaFold’s per-residue confidence, and sasa is the DSSP-derived solvent exposure of the residue. Both encode information about how exposed and well-modeled the residue is, and neither is recoverable from sequence alone. The explicit pocket-geometry features (dist_to_nearest_pocket, druggability, in_pocket) carry smaller individual importances but still contribute on top of pLDDT and SASA, which is why the structure-only arm reaches 0.74 macro-F1 rather than the 0.68 ceiling of sequence-only. The union’s win over single-modality arms is not driven by any one feature. It comes from combining three moderately strong signals from different sources.

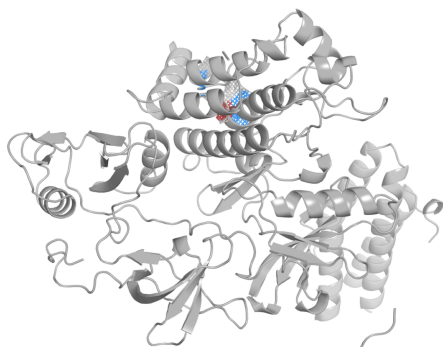
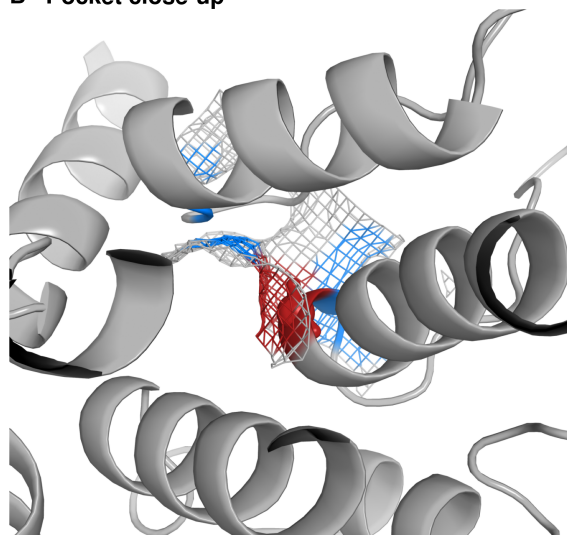
A Full protein**B Pocket close-up**

Figure 4: Case study, ABL1 p.A433T. **(A)** The variant residue (red) located within the AlphaFold-predicted ABL1 kinase fold; disordered tails (pLDDT < 50) are hidden for clarity. **(B)** A closeup of the residue inside its `fpocket`-predicted binding pocket (light-blue mesh), with pocket-lining residues drawn as blue sticks. On the test set the sequence-only XGBoost called this variant benign ($p = 0.22$); the full-feature model recovered the correct oncogenic call ($p = 0.89$). The sequence change A>T is modest on conventional substitution scores, but the structural context (deeply buried inside a druggable pocket, pLDDT 0.98) and high evolutionary conservation (phyloP 10.0) cause it to be flagged.

A second pattern in the headline table is that the gap between models shrinks as the feature set grows, which says the wins are feature-driven rather than model-driven. In all three models, the union of all feature sets yields a consistent +0.17–0.18 macro-F1 improvement, while the difference between models on the same feature set is less than 0.006 for the combined set and less than 0.019 for the individual feature sets, both of which fall within seed-to-seed noise. Model choice only mattered for the sequence-only feature set, where MLP underperformed tree-based models by about 0.01 macro-F1, suggesting that tree models are more robust when the feature set is thin.

Figure 4 shows what this looks like for a single variant. ABL1 is a non-receptor tyrosine kinase whose BCR-ABL fusion drives chronic myeloid leukemia and is the canonical target of imatinib, the first kinase inhibitor approved in oncology. Position 433 sits in the kinase domain, where the ATP-binding pocket is both catalytically essential and a known drug-binding site. ABL1 p.A433T is a variant in the test set that the sequence-only XGBoost called benign ($p = 0.22$) and the full-feature model rescued to a correct oncogenic call ($p = 0.89$). On BLOSUM62 alone the A to T swap is mild and so a model with nothing else to look at has no real reason to flag it. The full-feature model observes something different. The residue sits deep inside a druggable pocket (panel B), AlphaFold is confident in that placement (pLDDT 0.98), and the position is among most conserved (phyloP 10.0). Sequence said benign and structure plus evolution said oncogenic, and the latter is correct. One example does not generalize, but it shows the mechanism the headline numbers are picking up.

6 Conclusion

A three-way orthogonal ablation across sequence, structure, and evolutionary feature blocks shows that each modality carries non-redundant predictive signal for classifying missense variants as oncogenic versus benign. The full union beats every single-modality and pairwise arm in all three model classes, and every multi-modal combination beats the best of its constituents. Pocket-aware structural features are informative both on their own and inside the union, which supports the hypothesis that 3D context adds signal beyond what sequence- and conservation-based predictors already capture. Natural next steps include benchmarking against AlphaMissense as an explicit upper bound, replacing `fpocket` with P2Rank for pocket detection, and extending the feature block to include structural-aligned conservation (e.g. ConSurf-DB) on top of the per-base phyloP/phastCons signal.

Appendix A: Full results table

Table 1: Test split, mean $\pm$ std across 5 seeds. Bold = best feature set within each model (per metric); lower = better for ECE.

Feature set	Model	Macro F1	PR-AUC	ROC-AUC	Bal. Acc.	ECE
Sequence	Random Forest	0.684 $\pm$ 0.018	0.654 $\pm$ 0.041	0.747 $\pm$ 0.018	0.688 $\pm$ 0.020	0.074 $\pm$ 0.013
Sequence	MLP	0.674 $\pm$ 0.015	0.654 $\pm$ 0.041	0.746 $\pm$ 0.018	0.670 $\pm$ 0.015	0.030 $\pm$ 0.006
Sequence	XGBoost	0.686 $\pm$ 0.021	0.655 $\pm$ 0.041	0.748 $\pm$ 0.018	0.690 $\pm$ 0.021	0.075 $\pm$ 0.014
Structure	Random Forest	0.723 $\pm$ 0.034	0.718 $\pm$ 0.028	0.806 $\pm$ 0.037	0.721 $\pm$ 0.035	0.045 $\pm$ 0.028
Structure	MLP	0.740 $\pm$ 0.040	0.737 $\pm$ 0.034	0.818 $\pm$ 0.045	0.741 $\pm$ 0.043	0.050 $\pm$ 0.013
Structure	XGBoost	0.736 $\pm$ 0.032	0.731 $\pm$ 0.028	0.819 $\pm$ 0.036	0.740 $\pm$ 0.037	0.062 $\pm$ 0.014
Evolution	Random Forest	0.739 $\pm$ 0.020	0.700 $\pm$ 0.026	0.804 $\pm$ 0.022	0.740 $\pm$ 0.021	0.105 $\pm$ 0.014
Evolution	MLP	0.780 $\pm$ 0.019	0.761 $\pm$ 0.029	0.848 $\pm$ 0.017	0.781 $\pm$ 0.019	0.037 $\pm$ 0.005
Evolution	XGBoost	0.774 $\pm$ 0.017	0.774 $\pm$ 0.036	0.853 $\pm$ 0.019	0.779 $\pm$ 0.017	0.066 $\pm$ 0.005
Seq + Struct	Random Forest	0.813 $\pm$ 0.008	0.853 $\pm$ 0.011	0.895 $\pm$ 0.009	0.814 $\pm$ 0.008	0.027 $\pm$ 0.005
Seq + Struct	MLP	0.800 $\pm$ 0.016	0.841 $\pm$ 0.013	0.888 $\pm$ 0.011	0.799 $\pm$ 0.018	0.031 $\pm$ 0.011
Seq + Struct	XGBoost	0.812 $\pm$ 0.009	0.856 $\pm$ 0.009	0.897 $\pm$ 0.009	0.817 $\pm$ 0.009	0.041 $\pm$ 0.013
Seq + Evo	Random Forest	0.795 $\pm$ 0.012	0.817 $\pm$ 0.028	0.880 $\pm$ 0.013	0.798 $\pm$ 0.012	0.059 $\pm$ 0.010
Seq + Evo	MLP	0.809 $\pm$ 0.016	0.831 $\pm$ 0.025	0.894 $\pm$ 0.014	0.813 $\pm$ 0.017	0.023 $\pm$ 0.007
Seq + Evo	XGBoost	0.814 $\pm$ 0.013	0.847 $\pm$ 0.025	0.900 $\pm$ 0.013	0.823 $\pm$ 0.014	0.050 $\pm$ 0.007
Struct + Evo	Random Forest	0.819 $\pm$ 0.017	0.840 $\pm$ 0.022	0.899 $\pm$ 0.016	0.820 $\pm$ 0.017	0.022 $\pm$ 0.004
Struct + Evo	MLP	0.811 $\pm$ 0.022	0.831 $\pm$ 0.024	0.892 $\pm$ 0.020	0.813 $\pm$ 0.021	0.028 $\pm$ 0.010
Struct + Evo	XGBoost	0.815 $\pm$ 0.018	0.845 $\pm$ 0.019	0.900 $\pm$ 0.015	0.822 $\pm$ 0.018	0.043 $\pm$ 0.014
All Combined	Random Forest	0.863 $\pm$ 0.013	0.901 $\pm$ 0.016	0.936 $\pm$ 0.010	0.866 $\pm$ 0.011	0.030 $\pm$ 0.004
All Combined	MLP	0.857 $\pm$ 0.013	0.889 $\pm$ 0.017	0.930 $\pm$ 0.012	0.859 $\pm$ 0.011	0.020 $\pm$ 0.006
All Combined	XGBoost	0.861 $\pm$ 0.018	0.908 $\pm$ 0.011	0.939 $\pm$ 0.009	0.867 $\pm$ 0.016	0.034 $\pm$ 0.013

Appendix B: Full feature listing

The feature matrix has 54 columns across three orthogonal blocks (Table 2). The sequence block is purely a function of the wild-type and mutant residue identities and their physicochemical deltas; the structure block is per-residue output from AlphaFold + fpocket + mkdssp; the evolution block is per-nucleotide UCSC phyloP and phastCons.

Table 2: Per-column feature listing. AA binary-column flags are collapsed onto single rows (the 20 canonical amino acids). Block totals: sequence = 44, structure = 8, evolution = 2.

Block	Column	Description	Source
Seq.	Delta_Mass	MT minus WT average residue mass (Da).	Biopython <code>protein_weights</code>
Seq.	Delta_Hydro	MT minus WT Kyte–Doolittle hydropathy.	Biopython <code>ProtParamData.kd</code>
Seq.	Delta_Charge	MT minus WT net charge at pH 7 (R/K = +1, D/E = -1, H = +0.1, else 0).	Hand-curated from standard pKa tables
Seq.	BLOSUM62	WT to MT substitution score.	Biopython BLOSUM62
Seq.	WT_AA_* (20 cols)	20 binary columns, one per canonical AA; the column matching the WT residue is 1, others 0.	ClinVar Name parsing
Seq.	MT_AA_* (20 cols)	20 binary columns, one per canonical AA; the column matching the MT residue is 1, others 0.	ClinVar Name parsing
Struct.	dist_to_nearest_pocket	Minimum C α distance (Å) from the residue to the nearest pocket-lining residue.	fpocket
Struct.	in_pocket	1 if the residue itself lines any fpocket pocket, else 0.	fpocket
Struct.	druggability	Druggability score of the pocket whose nearest C α matched.	fpocket *_info.txt
Struct.	sasa	DSSP relative ASA (unitless, ~0–1; fraction of max-accessible surface area for that residue type).	mkdssp via Biopython DSSP
Struct.	plddt	AlphaFold per-residue confidence (0–100), read from the PDB B-factor column.	AlphaFold PDB
Struct.	ss_H / ss_E / ss_L (3 cols)	Three binary columns indicating secondary structure state (helix / strand / loop), collapsed from DSSP’s 8-state.	mkdssp
Evo.	phylop100way	Per-base substitution-rate deviation from a neutral model across the 100-vertebrate alignment (positive = conserved).	UCSC bigWig
Evo.	phastcons100way	Per-base posterior probability of belonging to a conserved element under the same alignment’s phylo-HMM (0–1).	UCSC bigWig

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